Variant Annotation with VEP

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Introduction

Ensembl's Variant Effect Predictor (VEP) assigns each variant a predicted consequence (missense, synonymous, splice-donor, intronic, etc.), gene and transcript IDs, and pathogenicity scores from predictors (SIFT, PolyPhen, CADD, REVEL). It is the standard annotator for variant interpretation in clinical genetics and population genomics.

Prerequisites

VCF file; Ensembl or RefSeq transcript database.

Theory

VEP traverses each variant, intersects it with transcript structures, and emits one record per overlapping transcript with the predicted functional consequence. Gene-level summaries pick the most severe consequence across transcripts (MANE-select or canonical transcript is standard for clinical use).

Assumptions

Reference build matches the VCF (GRCh37 vs GRCh38); transcript database is current.

R Implementation

VEP runs on the command line (Perl). For programmatic annotation in R, use `VariantAnnotation::locateVariants()` against a `TxDb` plus `predictCoding()` for protein consequences.

```r
library(VariantAnnotation)
library(TxDb.Hsapiens.UCSC.hg19.knownGene)
library(BSgenome.Hsapiens.UCSC.hg19)

vcf_file <- system.file("extdata", "chr22.vcf.gz", package = "VariantAnnotation")
vcf <- readVcf(vcf_file, "hg19")
seqlevelsStyle(vcf) <- "UCSC"

txdb <- TxDb.Hsapiens.UCSC.hg19.knownGene
loc  <- locateVariants(rowRanges(vcf), txdb, CodingVariants())
table(loc$LOCATION)

pc <- predictCoding(vcf, txdb, Hsapiens)
head(mcols(pc)[, c("CONSEQUENCE", "REFAA", "VARAA")])
```

A representative VEP command (not run here):

```
vep -i input.vcf -o output.vcf --cache --sift p --polyphen p --canonical
```

Output & Results

Per-variant, per-transcript annotation with consequence (missense_variant, synonymous_variant, etc.) and amino-acid substitution information.

Interpretation

"VEP annotation identified 12 missense variants predicted deleterious by SIFT and PolyPhen; one nonsense variant in BRCA1 was classified as likely pathogenic."

Practical Tips

- Always check assembly/transcript version match between caller and annotator.
- For clinical work, use VEP's `--pick` or `--flag_pick_allele_gene` to obtain a single canonical annotation per variant.
- Add pathogenicity plugins (CADD, REVEL, SpliceAI) for deleteriousness prediction.
- `ANNOVAR` is a competing annotator with a different gene-model approach; results can differ at splice boundaries.
- For batch annotation of millions of variants, VEP's offline cache mode is essential; REST/web queries do not scale.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- RNA-seq with DESeq2 / edgeR
- Enrichment analysis (GSEA / ORA)
- scRNA-seq with Seurat